

ThreadMatch – AI-Powered Adaptive Styling Engine

Project Overview

StyleAI is a full-stack web application designed to help users make smarter clothing decisions based on their body type, occasion, weather conditions, and personal style preferences.

The goal is to simulate the decision-making process of a personal stylist while providing outfit recommendations in seconds through an intuitive and interactive user experience.

This project was intentionally built as an MVP (Minimum Viable Product) to validate the core recommendation workflow before introducing advanced AI-generated styling and wardrobe intelligence features.

Problem Statement

Many users struggle with outfit selection despite owning multiple clothing items.

Common challenges include:

- Choosing outfits that fit a specific occasion
- Matching colors effectively
- Understanding which clothing silhouettes complement their body type
- Deciding what to wear based on weather conditions
- Lack of personalized styling guidance

Existing fashion applications focus primarily on shopping rather than helping users style the clothing they already own.

StyleAI focuses on recommendation and outfit composition rather than e-commerce.

Product Vision

Create an intelligent styling assistant capable of:

- Understanding user preferences
- Understanding wardrobe constraints
- Generating personalized outfit recommendations
- Explaining styling decisions
- Adapting recommendations dynamically

The long-term vision is to provide users with a virtual stylist available on demand.

MVP Scope

The MVP focuses on validating the recommendation workflow.

Included Features

<i>Authentication</i> • User Registration • User Login • JWT Authentication • Secure Password Hashing	<i>User Profiling</i> Users provide: • Gender • Body Shape • Occasion • Weather Context
<i>Wardrobe Builder</i> Users select: • Shirt Type • Pant Type • Preferred Shirt Colors • Preferred Pant Colors	<i>Recommendation Engine</i> The system generates: • Personalized outfit combinations • Color-based variations • Multiple recommendation options

User Flow

1. User creates an account
2. User signs in
3. User completes style questionnaire
4. User selects wardrobe preferences
5. Recommendation engine generates outfit combinations
6. User reviews personalized styling suggestions

System Architecture

Frontend:

- React, React Router, Tailwind CSS

Backend:

- Node.js, Express.js

Database:

- MongoDB Atlas, Mongoose ODM

Authentication:

- JWT, bcrypt

Deployment:

- Vercel (Frontend), Render / Railway (Backend)

Current Recommendation Logic

The MVP uses deterministic recommendation rules.

Examples:

- If a user selects a shirt only, recommend complementary pants.
- If a user selects pants only, recommend matching shirts.
- If a user selects both, generate color-based outfit variations.
- Multiple color selections generate multiple recommendation combinations.

This approach allows rapid iteration before introducing generative AI.

Future Roadmap

Phase 2

AI Recommendation Engine

Integrate OpenAI APIs to:

- Generate stylist explanations
- Create occasion-specific recommendations
- Produce confidence scores

Closet Management

Allow users to:

- Upload clothing inventory
- Manage wardrobes
- Track outfit usage

Visual Outfit Preview

Generate visual outfit mockups based on selected garments and colors.

Smart Weather Integration

Automatically adapt recommendations using live weather APIs.

Trend Intelligence

Introduce seasonal fashion recommendations and trend-based styling.

Status

Current Version: MVP v1.0

Project Status: Completed and actively evolving toward AI-assisted styling recommendations and wardrobe intelligence.